“AUTOMATED SEEDING BOT”

Getting started with IOT

SEP PROJECT REPORT

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INTRODUCTION

This project involves the design and implementation of an automated agricultural robot capable of performing a variety of tasks including drilling, seeding, flapping and watering. The robot is controlled using an Arduino microcontroller and features a combination of servo motors, DC motors, and sensors to carry out its functions. It can be operated manually through Bluetooth commands or run autonomously based on pre-defined sequences.

Functionality:

- **Movement:** The robot can move in multiple directions: forward, backward, left, and right. It can also stop and execute an emergency stop to halt all operations immediately.
- **Drilling:** The drilling function lowers and raises a drill mechanism using two synchronized servo motors, with a pump activated during the drilling process.
- **Seeding:** The seeder servo operates to plant seeds, with two modes available (seed and seed2) to handle different seeding requirements.
- **Flap Operation:** The flap servo controls an auxiliary flap mechanism, moving it through a full range of motion.
- **Irrigation:** Based on the readings from the moisture sensor, the pump can be activated to water the soil. The control logic ensures watering only when the soil moisture level falls below a specified threshold.
- **Automated Sequence:** The robot can run a complete automated sequence that integrates drilling, moving, seeding, flap operation, and watering, providing a fully autonomous agricultural cycle.

CODES

sketch_jun10c.ino

```
#include <Servo.h>

// Servo objects
Servo DRILL1;
Servo DRILL2;
Servo SEEDER;
Servo FLAP;

// Motor control pins
int m1 = 4;
int m2 = 5;
int m3 = 6;
int m4 = 7;

// Servo positions
int pos = 0;
int posf = 0;

// Motor pump control pins
int mp = 12;
int mn = 13;

// Moisture sensor pins
int pinD0 = 3;
int pinA0 = A0;
int motorPin = 2;

// Sensor values
int A0value = 0;
int D0value = 0;

// Incoming serial value
char Incoming_value = 0;

void setup() {
  // Start serial communication
```

sketch_jun10c.ino

```
void setup() {
  // Start serial communication
  Serial.begin(9600);


  // Attach servos
  DRILL1.attach(8);
  DRILL2.attach(9);
  SEEDER.attach(10);
  FLAP.attach(11);

  // Initialize servo positions
  DRILL1.write(0);
  DRILL2.write(180);
  SEEDER.write(0);
  FLAP.write(0);

  // Set pin modes
  pinMode(mp, OUTPUT);
  pinMode(mn, OUTPUT);
  pinMode(pinD0, INPUT);
  pinMode(pinA0, INPUT);
  pinMode(motorPin, OUTPUT);
  pinMode(m1, OUTPUT);
  pinMode(m2, OUTPUT);
  pinMode(m3, OUTPUT);
  pinMode(m4, OUTPUT);



  // Initialize Bluetooth module
  Serial.println("Bluetooth Ready");  // Signal that Bluetooth is ready
}

void loop() {
  // Read moisture sensor values1
```

sketch_jun10c.ino

```
// Function Definitions

void moveForward() {
  digitalWrite(m1, HIGH);
  digitalWrite(m2, LOW);
  digitalWrite(m3, LOW);
  digitalWrite(m4, HIGH);
  Serial.println("Moving Forward");
}

void moveBackward() {
  digitalWrite(m1, LOW);
  digitalWrite(m2, HIGH);
  digitalWrite(m3, HIGH);
  digitalWrite(m4, LOW);
  Serial.println("Moving Backward");
}

void turnRight() {
  digitalWrite(m1, LOW);
  digitalWrite(m2, LOW);
  digitalWrite(m3, HIGH);
  digitalWrite(m4, LOW);
  Serial.println("Turning Right");
}

void turnLeft() {
  digitalWrite(m1, LOW);
  digitalWrite(m2, HIGH);
  digitalWrite(m3, LOW);
  digitalWrite(m4, LOW);
  Serial.println("Turning Left");
}

```

```
void stopMovement() {
  digitalWrite(m1, LOW);
  digitalWrite(m2, LOW);
  digitalWrite(m3, LOW);
  digitalWrite(m4, LOW);
  Serial.println("Stopped");
}

void drill() {
  Serial.println("Drilling");
  for (pos = 0; pos <= 45; pos += 1) {
    DRILL1.write(pos);
    DRILL2.write(180 - pos);
    delay(5);
    digitalWrite(mn, LOW);
    digitalWrite(mp, HIGH);
  }
  delay(1500);
  for (pos = 45; pos >= 0; pos -= 1) {
    DRILL1.write(pos);
    DRILL2.write(180 - pos);
    delay(4);
    digitalWrite(mp, LOW);
    digitalWrite(mn, LOW);
  }
}
```

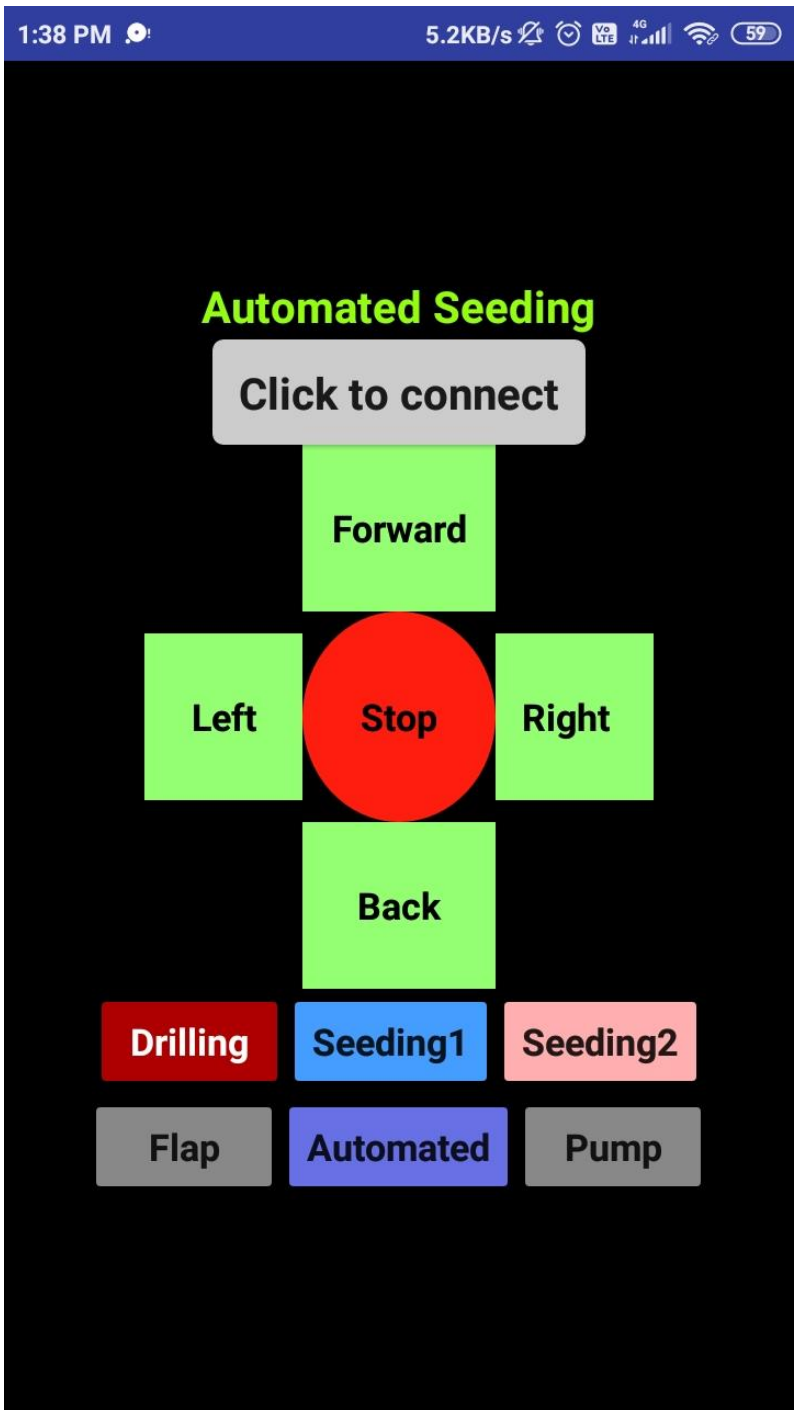
sketch_jun10c.ino

```
void controlPump() {
  // Serial.print("Moisture level: ");
  // Serial.print(A0value);
  // Serial.print(", Digital read: ");
  // Serial.println(D0value);


  Serial.println("Moisture level is 400 or above. Motor ON.");

  digitalWrite(motorPin, HIGH);
  delay(5000);
  digitalWrite(motorPin, LOW);
  Serial.println("CODE IS WORKING");

  // if (A0value < 400 && D0value == 0) {

  //   Serial.println("Moisture level is below 400. Motor OFF.");
  // } else {
  //   Serial.println("Moisture level is 400 or above. Motor ON.");

  //   delay(3000);
  //   digitalWrite(11, LOW);
  // }
}

void seed() {
  Serial.println("Seeding");
  SEEDER.write(0);
  delay(500);
  SEEDER.write(90);
  delay(500);
  SEEDER.write(0);
  delay(500);
}

```

```
void emergencyStop() {
  digitalWrite(m1, LOW);
  digitalWrite(m2, LOW);
  digitalWrite(m3, LOW);
  digitalWrite(m4, LOW);
  digitalWrite(motorPin, LOW);
  DRILL1.write(0);
  DRILL2.write(180);
  SEEDER.write(0);
  FLAP.write(0);
  Serial.println("Emergency Stop");
}

void automatedSequence() {
  Serial.println("Automated Sequence Start");

  drill();

  moveForward();
  delay(1350);

  stopMovement();

  seed();

  FLAP.write(35);

  controlPump();

  moveForward();
  delay(1500);
  stopMovement();
```

APP

when ListPicker1 .BeforePicking
do set ListPicker1 . Elements to BluetoothClient1 . AddressesAndNames
when ListPicker1 .AfterPicking
do set ListPicker1 . Selection to call BluetoothClient1 .Connect
address ListPicker1 . Selection
set ListPicker1 . Text to " CONNECTED "
when Forward .Click
do call BluetoothClient1 .SendText
text " 1 "
when right .Click
do call BluetoothClient1 .SendText
text " 3 "
when left .Click
do call BluetoothClient1 .SendText
text " 4 "
when Back .Click
do call BluetoothClient1 .SendText
text " 2 "
when stop .Click
do call BluetoothClient1 .SendText
text " 5 "
when seeding .Click
do call BluetoothClient1 .SendText
text " 8 "
when drilling .Click
do call BluetoothClient1 .SendText
text " 6 "
when flap .Click
do call BluetoothClient1 .SendText
text " 9 "
when pump .Click
do call BluetoothClient1 .SendText
text " 7 "
when seeding2 .Click
do call BluetoothClient1 .SendText
text " A "
when automated .Click
do call BluetoothClient1 .SendText
text " B "


1:38 PM
5.2KB/s
Automated Seeding
Click to connect
Forward
Left
Stop
Right
Back
Drilling
Seeding1
Seeding2
Flap
Automated
Pump

HARDWARE DEVICES

Arduino Microcontroller : The Arduino board is the central processing unit of the project. It runs the code that controls all other components and processes input from various sensors. It handles the logic for moving the robot, operating the servos, reading sensor data, and communicating via Bluetooth.

Servo Motors :

- **DRILL1 and DRILL2:** These servos control the drill mechanism. They are synchronized to move in opposite directions to lower and raise the drill.
- **SEEDER:** This servo operates the seeding mechanism, planting seeds in the soil.
- **FLAP:** This servo controls an auxiliary flap mechanism, which might be used for tasks such as covering seeds with soil.

DC Motors :

- **Movement Motors (m1, m2, m3, m4):** These four DC motors are used to drive the robot in various directions. By controlling the power to each motor, the robot can move forward, backward, and turn left or right.

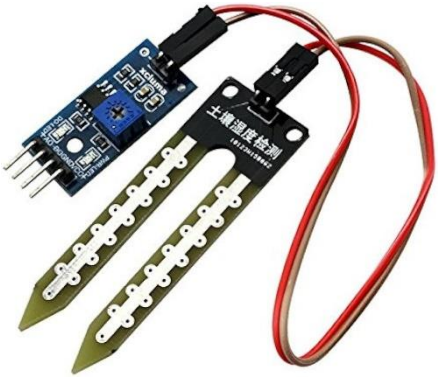
Pump:

The pump is used for irrigation. It is controlled via two pins (‘mp’ and ‘mn’). When activated, it pumps water to the soil based on the readings from the moisture sensor.

Moisture Sensor :

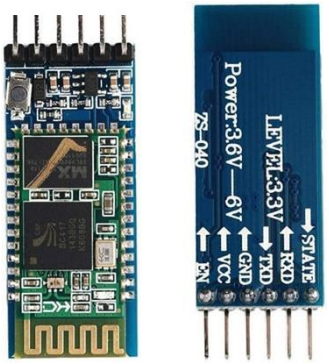
The moisture sensor has both analog and digital outputs:

- **<u>Analog Output (A0):</u>** Provides a continuous value representing the soil moisture level.
- **<u>Digital Output (D0):</u>** Indicates whether the soil moisture is above or below a certain threshold.



Bluetooth Module :

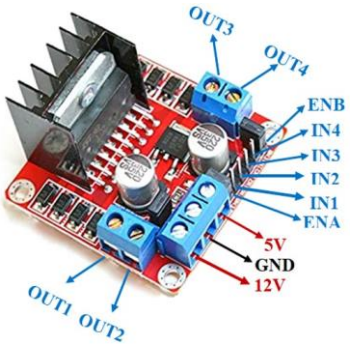
The Bluetooth module enables wireless communication with the robot. It allows for remote control and monitoring, sending commands to the Arduino and receiving status updates.



L298N motor driver :

The L298N motor driver is a robust and versatile dual H-Bridge motor driver that allows for controlling the direction and speed of two DC motors independently. It can handle high voltage and current, making it suitable for driving the four DC motors used in the agricultural robot for this project.

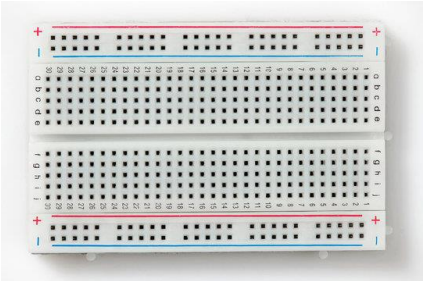
The L298N motor driver is a crucial component in the automated agricultural robot, providing reliable and efficient control over the DC motors. Its dual H-Bridge design allows the Arduino to control the movement of the robot with precision, enabling tasks such as drilling, seeding, and watering to be performed effectively. The ability to handle high currents and voltages makes it an ideal choice for driving the motors in this project, ensuring robust performance and durability.



Bread Board :

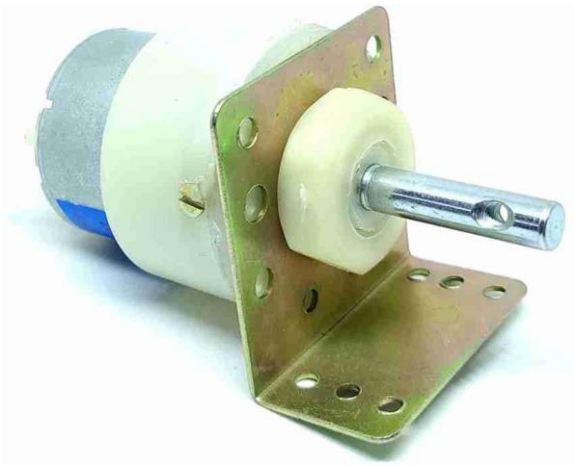
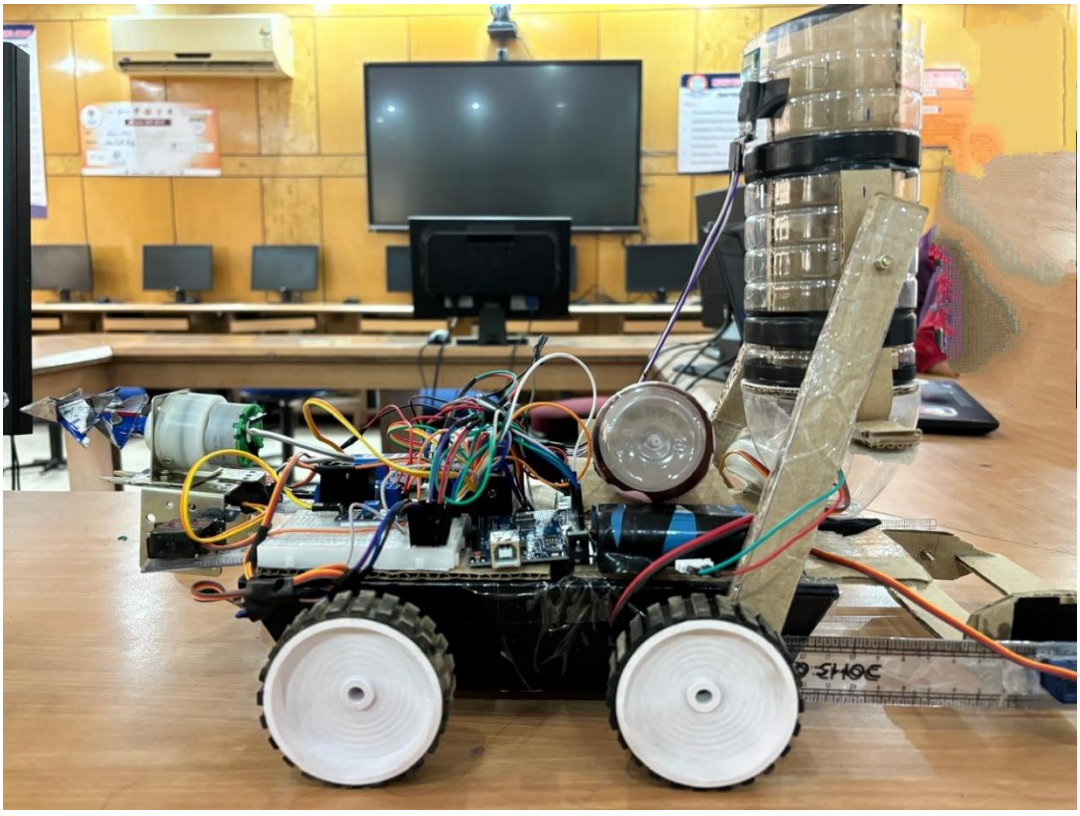
A breadboard is a tool for constructing electronic circuits without the need for soldering. It allows for easy and quick prototyping of electronic circuits by providing a platform to connect components using jumper wires. In this project, the breadboard is used to connect and organize the various components such as the Arduino, sensors, motors, and motor driver.

The breadboard is an essential tool in the agricultural robot project, enabling the assembly and testing of the electronic circuit without permanent soldering. It facilitates easy connections between the Arduino, L298N motor driver, sensors, servo motors, and other components, allowing for efficient prototyping and debugging. The use of a breadboard ensures that the circuit can be quickly modified and optimized, making it a vital part of the development process.



12v Motor Clamp :

A 12V motor clamp is a mechanical device used to securely mount DC motors in a variety of applications. In this project, the motor clamps are used to hold the 12V DC motors in place on the agricultural robot, ensuring they are securely attached to the chassis and properly aligned for optimal operation.



IMAGES